



RESEARCH ARTICLE

Shri Krishna R

Founder & Innovator

Phone No: 7845210170

Email: shrikrishnaravindran@gmail.com

AI-Based Real-Time Electrical Hazard Detection and Prevention System for Reducing Electrocution Deaths Caused by Snapped Power Lines and Current Leakage

Problem Statement

In villages and cities across Tamil Nadu, people continue to lose their lives due to electrical infrastructure failures. One of the most dangerous situations occurs when an overhead power line snaps and falls to the ground while still carrying live current. People, livestock, and wildlife may unknowingly come into contact with the conductor, resulting in severe electric shock, permanent injury, or death.

Current protection systems such as fuses, Miniature Circuit Breakers (MCBs), Ground Fault Protection (GFP), and other conventional protective devices do not always detect small breaks in branch conductors or partial wire snaps. As a result, a broken live conductor may remain energized for a significant period before authorities become aware of the hazard.

Even a single unnoticed wire break can become life-threatening.

Analysis of Electrocution Incidents

According to publicly available data, between 2001 and September 2023:

- 10,718 electrocution-related deaths were reported.

- 2,041 deaths were linked to overhead conductors.
- 324 non-fatal human injuries were recorded.
- 89 deaths occurred during 2022–2023 alone.
- 2,495 animals, including domestic and wild animals, were killed due to electrical infrastructure failures over the last seventeen years.

Source:<https://www.dtnext.in/news/tamilnadu/pil-seeks-tangedco-to-provide-earthing-to-all-overhead-lines-798889>

Recent incidents further highlight the severity of the issue:

- July 2, 2025 – Chennai (Tiruvottiyur)

A 17-year-old boy died after stepping into water that had become electrically energized due to infrastructure failure.

Source:<https://www.indiatoday.in/cities/chennai/story/chennai-teen-dies-of-electrocution-locals-blame-power-board-say-complaints-were-ignored-2750049-2025-07-03>

- July 24, 2025 – Kilambakkam, Chennai

An auto driver was electrocuted after a high-tension wire snapped and fell.

Source:<https://www.newindianexpress.com/states/tamil-nadu/2025/Jul/24/chennai-auto-driver-electrocuted-as-high-tension-wire-snaps-near-kilambakkam>

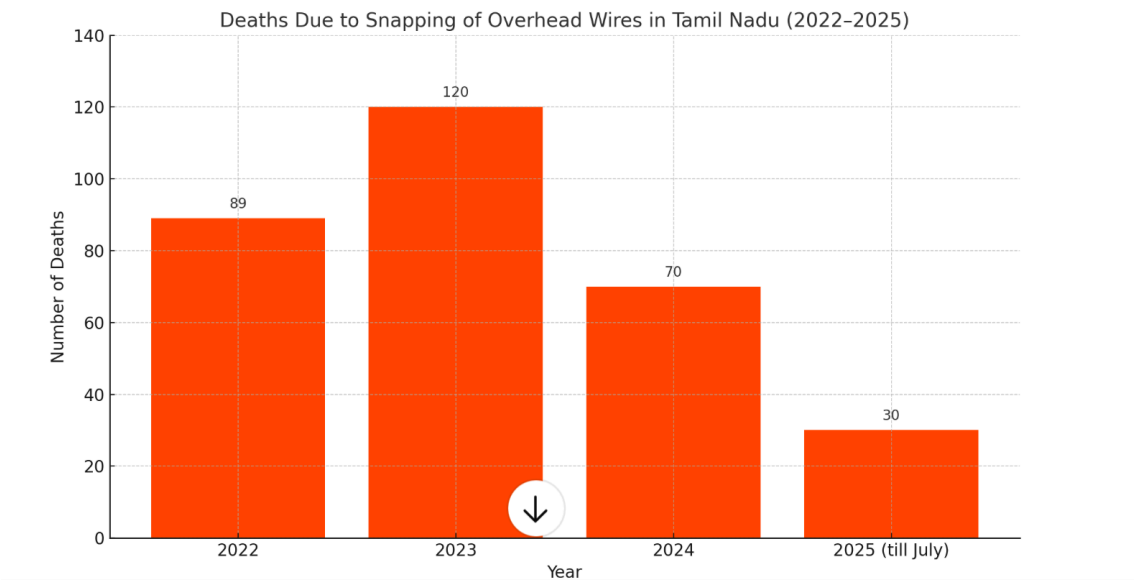
- August 10, 2025 – Tambaram

A man was electrocuted due to electrical infrastructure failure.

Source:<https://www.newindianexpress.com/states/tamil-nadu/2025/Aug/13/man-electrocuted-near-tambaram>

Category	Data
Total electrocution deaths (2001 – Sep 2023)	10,718
Deaths caused by overhead conductors	2,041
Non-fatal human injuries	324
Deaths during 2022–23	89
Animal deaths (domestic and wild animals)	2,495

Date	Location	Incident Description
02 Jul 2025	Tiruvottiyur, Chennai	17-year-old boy electrocuted due to electrical leakage associated with infrastructure failure and waterlogging
24 Jul 2025	Kilambakkam, Chennai	Auto driver electrocuted after a high-tension wire snapped and fell
10 Aug 2025	Tambaram, Chennai	Man electrocuted due to electrical infrastructure failure
22 Aug 2025	Chennai	57-year-old man electrocuted after contact with a snapped cable
07 Mar 2026	Madurai	20-year-old youth and his uncle died after encountering a live electrical conductor



J. Radhakrishnan Special Committee

Timeline:

- 2 July 2025 → Electrocution death reported in Tiruvottiyur.
- 3 July 2025 → Special committee constituted under Dr. J. Radhakrishnan.
- 18 July 2025 → Original deadline for submission of committee findings.

Following the committee's formation:

- TNPDCCL identified 27 exposed underground cable locations and rectified them.
- Later inspections identified 674 exposed underground cables.
- Approximately 80% of these locations were repaired as part of safety and pre-monsoon measures.

Source:<https://timesofindia.indiatimes.com/city/chennai/tnpdcl-works-on-fixing-exposed-ug-power-cables-as-residents-seek-rectification/articleshow/123487213.cms>

However, despite these interventions, Chennai recorded five electrocution deaths within approximately two months, indicating that the underlying issue continued to persist.

Source:<https://www.newindianexpress.com/cities/chennai/2025/Aug/24/on-the-danger-line-chennai-records-five-electrocution-deaths-ahead-of-northeast-monsoon>

- 1.The committee and subsequent investigations highlighted an important operational challenge:
- 2.Officials from TNPDCCL stated that underground power cables are often damaged during road cutting, utility excavation, drainage work, and infrastructure projects. In many cases, TNPDCCL is not informed when such damage occurs.
- 3.Repeated meetings between TNPDCCL, Chennai Corporation, Metro Water, and other departments have not completely eliminated communication gaps.
- 4.This demonstrates that administrative coordination alone may not be sufficient to prevent future electrocution incidents.

Date	Event
02 Jul 2025	Electrocution death reported in Tiruvottiyur
03 Jul 2025	Special committee constituted under Dr. J. Radhakrishnan
18 Jul 2025	Original deadline for submission of committee findings
Aug 2025	TNPDCCL reported identification of 27 exposed underground cable locations and rectification
Aug 2025	TNPDCCL identified 674 exposed underground cables, approximately 80% repaired

Why Are We Not Able to Prevent These Deaths?

- 1.Outdated Infrastructure
- 2.No Real-Time Monitoring
- 3.Lack of Automated Isolation

- Environmental Conditions
- Delayed Hazard Identification
- Inter-Departmental Coordination Issues

Objectives

- Prevent deaths and injuries caused by snapped live conductors.
- Detect even minor conductor failures before they become fatal hazards.
- Provide real-time alerts to authorities and emergency response teams.
- Enable rapid fault isolation and hazard mitigation.
- Develop a low-cost and scalable solution that can be deployed throughout existing power distribution infrastructure.

Existing Systems

- RCD / RCCB / GFCI (Residual Current Device)

Used in countries such as Germany and China to detect leakage currents and disconnect power.

- Broken Conductor Detection (BCD) Relays

Used in Japan and the United States to identify conductor failures.

- Phasor Measurement Unit (PMU) Based Systems

Used by utilities such as PG&E in California for grid monitoring and wildfire prevention.

- Ground Fault Protection (GFP)

Widely used across many countries to detect leakage currents.

Limitations of Existing Systems

- Aging infrastructure remains largely unmonitored.
- Advanced systems are expensive for widespread deployment.
- Existing technologies focus on substations rather than public safety hazards at street level.
- Many solutions are difficult for field personnel to install and maintain.
- Conventional protection mechanisms often fail to identify low-current conductor snaps.
- Most systems react to faults rather than predicting them

Critical Deployment Zones

The proposed solution is most urgently required in major urban centers where population density and electrical infrastructure complexity are highest.

Priority locations include:

1. Chennai
2. Coimbatore
3. Tiruchirappalli
4. Salem

Among these, Chennai remains the highest priority due to the concentration of reported electrocution incidents and recurring infrastructure-related hazards

Proposed Solution

- To address the limitations of existing protection systems, a new intelligent framework named Adaptive Electrical Hazard Intelligence Network (AEHIN) is proposed.
- The system introduces an AI-driven approach for the early identification and prevention of electrocution hazards before human contact occurs.

Algorithm Overview

Step 1: Continuous Infrastructure Observation

Step 2: Hazard Pattern Analysis

Step 3: Risk Classification

Step 4: Predictive Hazard Assessment

Step 5: Automated Alert Generation

Step 6: Continuous Learning

Expected Impact

1. Significant reduction in electrocution deaths.
2. Early identification of hazardous electrical conditions.
3. Faster response by utility authorities.
4. Improved public safety during monsoon seasons.
5. Protection of humans, livestock, and wildlife.
6. Reduction in compensation costs and emergency response burdens.
7. Creation of a proactive rather than reactive electrical safety ecosystem.

AI-Based Electrical Hazard Detection and Prevention Algorithm

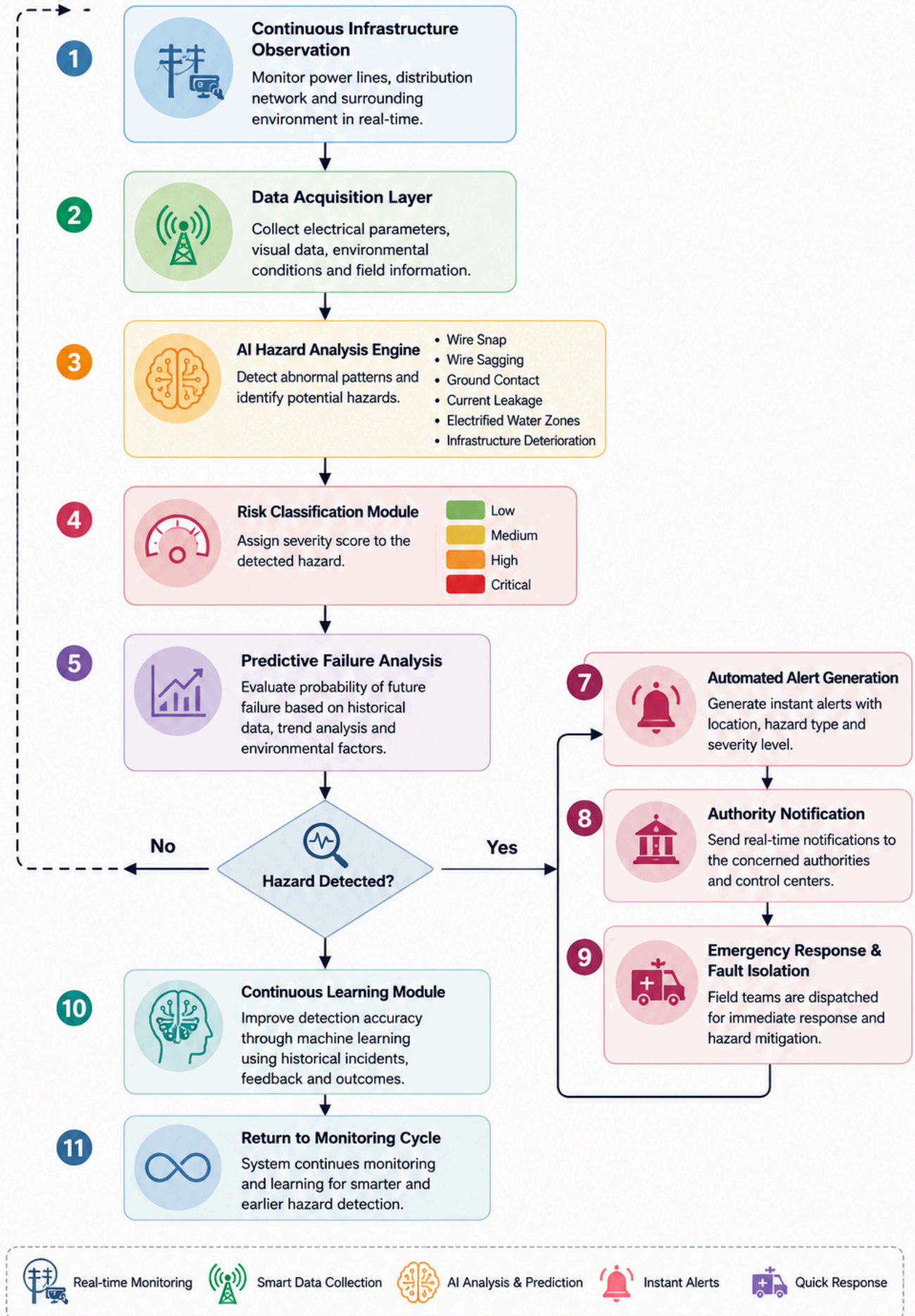


Figure 1: Workflow of the AI-Based Electrical Hazard Detection and Prevention Algorithm

Conclusion

The recurring electrocution incidents reported across Tamil Nadu demonstrate that existing protection mechanisms and administrative interventions alone are insufficient to eliminate the risk posed by snapped conductors and electrical leakage. The proposed Adaptive Electrical Hazard Intelligence Network introduces a proactive, AI-driven framework capable of detecting, analyzing, predicting, and reporting hazardous electrical conditions before they result in injury or loss of life. By shifting from fault response to hazard prevention, the system has the potential to significantly improve public electrical safety across both urban and rural environments.